

Activity 10

Md. Ahsan Kabir
0122520016

A/B experiment for Prothom Alo Homepage.

Driving Question: How can we increase user engagement on the Prothom Alo homepage?

Hypothesis: Redesigning the homepage to prioritize mobile-friendly, card-based layouts with larger images & shorter headlines will increase average session duration by 15% (maybe), as it improves readability and scannability on mobile devices, where most traffic originates.

Experiment Design:

- Variants:
 - Variant A (Control): Current homepage with traditional list-style news feed.
 - Variant B (Test): Card-based layout with larger thumbnails, truncated headlines (max 10 words) and infinite scroll.
- Key Performance Indicator (KPI): Average session duration (tracked by Google Analytics).
- Setup: Randomly split 50% of users to each variant; target new visitors to avoid bias; run on desktop and mobile.

Explanation of Choices : The card-based layout for Variant B was chosen because Prothom Alo's traffic is $\sim 70\%$ mobile (based on industry benchmarks for news sites in Bangladesh). Cards enhance touch navigation & visual appeal, reducing bounce rates compared to dense lists.

Experiment Duration : 2 weeks, to collect data from at least 10,000 sessions per variant (assuming daily traffic of ~ 1 million uniques), ensuring statistical significance (95% confidence) while avoiding weekend/weekday biases.

Hoped Results : Variant B will show a statistically significant increase in average session duration (e.g. from 2:30 to 2:50 minutes) with lower bounce rates, confirming the hypothesis & justifying a full rollout.